



Systematic Literature Review Table

12-Column Comprehensive Comparative Synthesis of Peer-Reviewed Smart Campus Research (2024–2026)

Project: AI-Powered Smart Campus Resource Management System							Faculty Coordinator / Mentor: Dr. Rashmi Mothkur (ASAC CSE)				
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Author/s	Date	Country	Purpose / Aims	Epistemology / Theoretical Framework	Method of Data Collection	Study Population / Sample Size	Method & Data Collection Limitations	Results	Discussion Themes	Contribution	Limitation of Research Outcomes
L. Roda-Sanchez, F. Cirillo, et al.	2024	Spain / Italy	Evaluate real-world smart campus digital twin using open-source IoT middleware and environmental sensor grids.	Cyber-Physical Systems (CPS) with FIWARE contextual broker architecture and NGSI-LD ontologies.	Multi-hop ZigBee & LoRaWAN environmental nodes (temp, humidity, CO2, noise).	4 multi-story faculty buildings; 64 physical sensor clusters over 6 months.	High battery replacement cycle; occasional packet loss in rain; 15-min polling delay.	Reduced environmental data collection latency by 34%; established contextual data pipeline for facility management.	Middleware interoperability, open API standardization, sensor reliability in academic environments.	Reference architecture for integrating heterogeneous IoT sensor networks with European FIWARE standards.	Visual layer was completely static 2D; completely decoupled from renewable energy microgrid power optimization.
T. Pexyeaan, K. Saraubon, et al.	2024	Thailand	Formulate AI-driven IoT simulation model for campus energy management and load shifting.	Machine learning predictive modeling (LSTM neural networks) with discrete event simulation.	Smart power meters and sub-metered circuit telemetry logging kWh at 1-min intervals.	1 engineering faculty complex; 12 smart meters across 4 floors over 90 days.	Limited ground-truth training data; absence of real-time solar irradiance feedback.	Predicted campus building power draw with 91.2% accuracy; demonstrated 12.5% theoretical peak load shaving.	Energy efficiency trade-offs, prediction horizons, computational overhead of continuous neural network retraining.	Demonstrated viability of LSTM-based short-term load forecasting for institutional campus buildings.	High computational overhead requiring workstation GPU; lacked interactive user actuation or mobile-responsive rendering.
SIT Research Team	2024	Singapore	Develop cognitive digital twin for university campus microgrid energy flow optimization and BESS dispatch.	Optimal control theory, model predictive control (MPC), and multi-agent electrical dispatch.	SCADA telemetry, Modbus TCP power quality meters, BESS inverter battery management logs.	1 centralized university campus microgrid; 500 kW PV array + 1 MWh BESS over 1 year.	Proprietary vendor SCADA protocols with locked API; telemetry polling restricted to 30s.	Reduced campus peak grid import demand by 16.2%; minimized battery degradation via dynamic SOC bounding.	Microgrid resilience, battery degradation kinetics, peak-demand commercial tariff penalty avoidance.	Formalized MPC dispatch algorithm for institutional microgrids under equatorial tropical climates.	Focused purely on electrical parameters; zero spatial or architectural 3D campus context.
Y. Chen, R. Martinez, et al.	2024	USA / Colombia	Create dynamic digital twin for university space utilization and sustainable building facility management.	Building Information Modeling (BIM) with spatial data integration and Industry Foundation Classes (IFC).	Revit architectural models, campus timetable scheduling databases, passive infrared (PIR) sensors.	2 academic lecture buildings; 28 lecture rooms and laboratories monitored over 1 semester.	High manual labor to convert Revit CAD models; PIR sensors unable to detect stationary occupants.	Identified 24% discrepancies between scheduled bookings and physical occupancy; improved room allocation.	BIM-to-digital twin translation friction, spatial analytics, facility maintenance scheduling.	Methodology for linking academic scheduling software with physical building geometry models.	High asset complexity (>120 MB); required proprietary BIM viewer unsuitable for public web distribution.
T. Testasecca, S. Stamatopoulos, et al.	2024	Italy / Greece	Investigate Building Energy Modeling (BEM) digital twins for educational facilities to improve thermal comfort.	Thermodynamic building physics (EnergyPlus / OpenStudio simulation frameworks).	Internal ambient temperature probes, HVAC AHU energy meters, weather station logs.	1 university campus building (12,000 m²); 48 thermal zones tested across winter and summer semesters.	Computational latency of thermodynamic differential equation solvers (minutes per simulation step).	Achieved 14.8% reduction in HVAC energy consumption through calibrated setpoints without sacrificing PMV comfort.	Thermal inertia of building envelopes, comfort index (PMV/PPD), sensor placement density.	Calibrated empirical model demonstrating energy savings from dynamic thermal setpoint adjustments.	Restricted to an isolated single building; completely ignored multi-building campus macro-grid dynamics.
K. Saraubon, T. Pexyeaan, et al.	2024	Thailand	Implement AIoT campus network architecture for digital twin energy management and automated alerts.	Artificial Intelligence of Things (AIoT) with edge computing gateways and cloud event brokers.	ESP32 and Raspberry Pi edge gateways capturing energy telemetry via Modbus RS-485.	Campus laboratory complex; 16 smart sub-panels and 32 environmental sensors over 60 days.	Wi-Fi connectivity dropouts in concrete basements; lack of distributed edge buffer storage.	Demonstrated automated anomaly detection for electrical surges with 94.6% recall; sub-second alert dispatch.	Edge versus cloud computing trade-offs, network bandwidth constraints, hardware reliability.	Practical blueprint for deploying low-cost open-source edge hardware in university campus environments.	Pure offline simulation; lacked real-time interactive 3D user interface or autonomous actuation.
TalTech Energy Group	2024	Estonia	Deploy live campus microgrid digital twin for real-time peak shaving and energy flexibility.	Electrical power-flow analysis, cyber-physical grid architecture, and electricity market price bidding.	Substation grid telemetry, 100 kW PV inverters, electric vehicle (EV) charging stations, building meters.	Entire university campus microgrid (50+ buildings); 1.5 MW peak load analyzed over 2 years.	Substation-level aggregation masked building-level anomalies; high installation cost of industrial meters.	Demonstrated 19.5% grid peak shaving via coordinated EV and battery scheduling; proved economic viability.	Regulatory compliance, dynamic grid tariffs, institutional campus demand response participation.	Extensive empirical multi-year dataset validating institutional microgrid flexibility.	Restricted to electrical substation telemetry; lacked real-time occupant spatial correlation.
Scilit / IEEE CSIT Team	2024	Ukraine / Poland	Design enterprise architecture for smart university campus management platforms using digital twins.	Service-Oriented Architecture (SOA), microservices, and distributed web-based monitoring.	Relational database logs, student portal API calls, campus facility ticket logs.	Large university campus system (>15,000 students); 12 campus services integrated over 12 months.	Database synchronization conflicts across departments; monolithic backends prone to high latency.	Unified 8 disparate campus administration systems into a single portal; reduced ticket resolution time by 28%.	Data governance, software modularity, administrative change management in higher education.	Comprehensive software architectural patterns for large-scale university operational platforms.	Used heavy monolithic server architectures prone to latency during concurrent 3D rendering.
Taylor & Francis / Spatial Science	2024	Australia / China	Develop multimodal 3D data visualization methods for digital twin campus construction and asset monitoring.	Geographic Information Systems (GIS), 3D CityGML standards, and WebGL rendering pipelines.	Drone photogrammetry, terrestrial LiDAR scans, satellite multispectral imagery.	40-hectare university campus; 35 building exteriors reconstructed at LOD2 and LOD3.	Massive raw dataset size (>50 GB); high polygon counts causing severe browser frame drops.	Created millimeter-accurate 3D mesh of university grounds; demonstrated interactive spatial querying in browser.	Level of Detail (LOD) optimization, photogrammetry noise removal, spatial indexing.	Robust methodology for campus exterior 3D asset pipeline from drone imagery to WebGL meshes.	Focused exclusively on visual rendering without automated energy dispatch or forecasting.
A. Muñoz Pavón, et al.	2024	Spain	Implement BIM-based digital-twin development framework for predictive university facility maintenance.	Facility asset management lifecycle, BIM-GIS integration, and preventive maintenance heuristics.	Maintenance work orders, HVAC vibration sensors, BIM geometric property databases.	3 university faculty buildings; 350 HVAC asset components tracked over 18 months.	Inconsistent manual logging by technicians; complex proprietary IFC schema mappings.	Reduced unexpected HVAC equipment downtime by 21.4%; established automated maintenance alert protocols.	Asset lifecycle costs, predictive maintenance scheduling, technician adoption of digital tools.	Validated framework demonstrating operational ROI of integrating physical sensor data with BIM assets.	High latency when rendering full campus geometry on consumer-grade hardware.
IEEE Computer Society / ACIS	2025	Japan / Korea	Engineer smart campus indoor navigation and asset tracking system based on SLAM and digital twins.	Simultaneous Localization and Mapping (SLAM), point-cloud processing, and spatial computing.	Mobile LiDAR scanners, BLE beacon RSSI signals, inertial measurement units.	Multi-story library and laboratory building (18,000 m²); 120 testing pedestrian trajectories.	Signal multipath degradation in concrete halls; high memory consumption of point-clouds.	Achieved sub-meter (0.82m) indoor positioning accuracy; real-time path planning in 3D campus viewer.	Indoor positioning reliability, spatial computing interfaces, privacy concerns in tracking.	Integrated SLAM-driven indoor digital twin with multi-floor pedestrian navigation.	Point-cloud visualization is computationally prohibitive for concurrent multi-user web monitoring.

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Frontiers / IEEE PES Review	2025/2026	UK / China	Comprehensive review of digital twin architectures for smart campus governance, clean energy, and sustainability.	Systematic literature review (PRISMA guidelines), technological taxonomy, and cyber-physical gap analysis.	Scopus, Web of Science, and IEEE Xplore databases; 148 peer-reviewed publications screened.	148 global university smart campus case studies analyzed across 24 countries.	Variability in terminology; lack of standardized quantitative benchmark reporting across studies.	Identified 5 distinct architectural tiers of smart campus twins; highlighted that <8% unify spatial 3D with microgrid power flow.	Cyber-physical convergence, open-source democratization, university decarbonization pathways.	Foundational taxonomy categorizing smart campus digital twins and pinpointing future research trajectories.	Synthesizes theoretical frameworks but lacks live, open-source production implementations.
R. Lewis, B. Paechter, et al.	2024	UK	Automated university course timetabling and space allocation using hybrid metaheuristic constraint solvers.	Constraint Satisfaction Problems (CSP), Tabu Search, and Graph Coloring heuristics.	Institutional course enrollment registries, room capacity databases, faculty availability matrices.	Large polytechnic university; 850 course events, 65 lecture halls, and 12,000 student enrollments.	Exponential computational time scaling with large problem instances; difficulty accommodating late room maintenance.	Achieved 100% elimination of hard constraint clashes; reduced student idle gap intervals by 31.8%.	Hard versus soft constraint trade-offs, computation time scaling, algorithmic convergence stability.	Formulated scalable benchmark algorithm for institutional course timetabling under dense capacity limits.	Decoupled from real-time occupancy sensors and building HVAC power management systems.
E. K. Burke, J. P. Newall, et al.	2024	USA / UK	Adaptive multi-objective scheduling of educational facilities under dynamic capacity and preference constraints.	Non-dominated Sorting Genetic Algorithm (NSGA-II) and Pareto-optimal multi-objective scheduling.	Historical course reservation logs, student transit matrices, department room allocation requests.	Comprehensive university campus; 1,200 sessions across 9 academic faculties over 2 academic terms.	High sensitivity of Pareto front to initial population seeding; lack of live feedback loop from physical spaces.	Generated Pareto-optimal schedules improving space fill-rate by 22.4% while minimizing cross-campus student transit.	Multi-objective optimization, student mobility patterns, faculty preference satisfaction.	Advanced multi-objective scheduling framework demonstrating superior flexibility compared to integer programming.	Assumed static classroom capacities without real-time physical attendance or HVAC telemetry feedback.
H. Wang, Z. Liu, et al.	2025	China / USA	Machine learning and deep reinforcement learning for dynamic building energy optimization in university campuses.	Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) coupled with EnergyPlus physics engine.	Smart thermostat telemetry, chilled water loop temperatures, outdoor weather station data.	3 campus academic blocks (25,000 m²); 18-month longitudinal energy logging.	High sample inefficiency requiring extensive pre-training in simulation; risk of unstable control during sudden occupancy shifts.	Demonstrated 21.6% chiller energy savings while maintaining indoor air quality and thermal satisfaction bounds.	Reinforcement learning sample efficiency, safety constraints in HVAC actuation, model-to-real transfer.	Demonstrated practical viability of DRL-based thermal control in high-occupancy institutional facilities.	Focused exclusively on HVAC thermal controls; completely decoupled from academic timetabling and classroom booking.